

The build is the easy half. Trust is the product.

A realistic support flow, the controls around it, and the LangSmith loop that caught a fluent failure no error dashboard could see.

THE RELIABILITY PROBLEM

“I’m handing this to Steve... I’ve passed along your concern.”

— Tool calls: none

Warm tone. Correct facts. Green trace. The customer waits for help that was never requested.

BUILT HERE

LangChain

Fast, configurable agent harness. Six tools and middleware on one standard loop.

RUNTIME UNDERNEATH

LangGraph

Durable state, explicit topology, streaming, checkpointing, and human interrupts.

GRADUATE WHEN NEEDED

Deep Agents

Long-horizon planning, subagents, skills, and a virtual filesystem.

LangSmith

☐ Observe

☐ Evaluate

☐ Improve

Business hypothesis — resolve billing accurately, then use the resolved moment for discovery. Recommendation attach rate is unknown until real traffic.

Let the model handle language. **Never identity or money.**

The agent chooses a bounded capability. Deterministic layers decide whose data it can touch, perform arithmetic, and control side effects.

SERVER**Credential → customer principal**

The request body cannot choose the tenant. A mismatch is refused before a run exists.

RUNTIME**Frozen AuthContext**

Identity is out-of-band: not chat text, state, or a model-visible tool argument.

DATA**Repository is permanently scoped**

Parameterized SQL binds the tenant; Chinook opens read-only; other-tenant misses reveal nothing.

DEFENSE**Audit + result guard**

A future tool that crosses the boundary turns a silent leak into a loud failed turn.

FLAT CREATE_AGENT · SIX TOOLS

One loop, bounded capabilities

BILLING · 3 List invoices · invoice detail · SQL spend summary

DISCOVERY · 1 Catalog-grounded recommendations excluding owned tracks

SUPPORT · 2 File refund request · queue human handoff

Refund request

Human approval + stable idempotency key

Handoff

Durable row, status queued—not sent

PRECISE CLAIM

Production callers: credential binding + tenant-bound threads + scoped queries block the tested unauthorized reads and writes.

Not claimed: the model is unfoolable or every sentence is true. That is a different control loop.

Complexity did not earn its hop. Flat ships.

Same model, same 30-example dataset, same evaluators, same commit. The supervisor had to clear every pre-registered bar.

SHIPS

Flat

Resolved	30 / 30
p50 full turn	3.8s
Tool calls	45
Security failures	0

CHALLENGER

Supervisor

Resolved	30 / 30
p50 full turn	6.6s
Tool calls	61
Security failures	0

WHAT MADE THE DECISION CREDIBLE

The first controlled freeze showed 29–30. I did not bank a one-example lead.

The bar was written before the challenger existed. After the escalation fix, both arms were rerun at the same commit and tied. What survived was the extra hop: 1.7× median latency and 16 more tool calls.

Decision rule: visible quality gain + no workflow regression + $\leq 2.0s$ latency penalty. Ties go to the simpler topology.

PROVENANCE

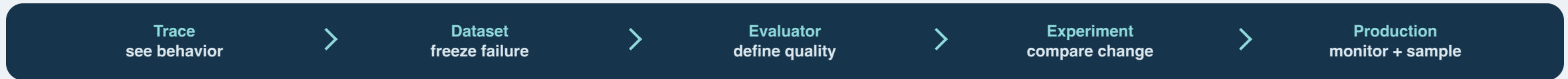
commit a29a530 · clean

dataset 6ba144216f8d

evaluators decda2664959

A trace proves a call happened. **Not that the promise is true.**

This single handoff failure needed four levels of evidence. Each correction became input to the next part of the engineering loop.



State-aware replay — clone a traced thread back into the local graph.

Trace → dataset — turn a real failure into a permanent regression case.

One quality definition — reuse evaluators offline and online.

Pilot two workflows. Stop on two failures.

The build is ready for a bounded evaluation—not a production victory lap. The missing numbers require real customers and real reviewers.

Measured on this build

Task resolution, frozen comparison	30 / 30
Cross-tenant access / unapproved refund writes	0 / 0
Flat p50 full-turn latency	3.8s
Deterministic pre-flight	139 passed

Unknown until production traffic

Containment without unsafe resolution	unknown
Recommendation attach / conversion rate	unknown
Human review agreement and downstream outcome	unknown
Actual queue shape and integration effort	unknown

FRICION
WORTH FIXING

Authorization payload hid run context under kwargs; the obvious check failed open.

Useful HITL cards required async prefetch because the description callback cannot do I/O.

Middleware composed a prompt no UI exposed—the root cause of the fabricated handoff.

TWO-WEEK PILOT

Scope
billing lookup + refund intake

Measure
containment + unsafe-resolution

Safety stop
any foreign-tenant access or unapproved refund write

Next: add support-rep annotation queue; hold recommendations behind an attach-rate experiment.

The build is not the bottleneck. Getting to reliable is.